MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC TEST – 04 (POLYNOMIALS)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 2: Polynomials** | **MM: 30**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, zero skipped steps, and final boxed answers with proper units.
2. In alternative (**OR**) sub-questions, complete step-by-step solutions and equal step marking apply to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Roots rejected by physical domain/restrictions are explicitly stated.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective & MCQs

MATHS MANTHAN | $5 \times 1 = 5$ Marks

Q1. If α and β are zeroes of $x^2 - 2x - 15$, then the polynomial whose zeroes are 2α and 2β is: [1]

(A) $x^2 - 4x - 60$ (B) $x^2 + 4x - 60$ (C) $x^2 - 4x + 60$ (D) $x^2 - 8x - 30$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

From $x^2 - 2x - 15$, $\alpha + \beta = 2$ and $\alpha\beta = -15$. We know:

Formula Used: $p(x) = k\left[x^2 - S'x + P'\right], \quad S' = 2(\alpha + \beta),\ P' = 4\alpha\beta$

Thus $S' = 2(2) = 4$ and $P' = 4(-15) = -60$, giving $\boxed{x^2 - 4x - 60}$. [1M]

Q2. If α, β are zeroes of $x^2 - 3x - 2$, then the value of $\alpha^3 + \beta^3$ is: [1]

(A) 45 (B) 27 (C) 36 (D) 18

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Here $\alpha + \beta = 3$ and $\alpha\beta = -2$. We know:

Formula Used: $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

Thus $\alpha^3 + \beta^3 = 3^3 - 3(-2)(3) = 27 + 18 = \boxed{45}$. [1M]

Q3. If the zeroes of $f(x) = 2x^2 + 5x - 12$ are α and β, then $\alpha\beta^2 + \alpha^2\beta$ is equal to: [1]

(A) 15 (B) -15 (C) 30 (D) -30

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

From $2x^2 + 5x - 12$, $\alpha + \beta = -\frac{5}{2}$ and $\alpha\beta = \frac{-12}{2} = -6$. We know:

Formula Used: $\alpha\beta^2 + \alpha^2\beta = \alpha\beta(\alpha + \beta)$

Therefore $\alpha\beta(\alpha + \beta) = (-6)\left(-\frac{5}{2}\right) = \boxed{15}$. [1M]

Q4. If α, β are the zeroes of $f(x) = x^2 - 4$, then $\alpha^4 + \beta^4$ is: [1]

(A) 32 (B) 16 (C) 64 (D) 0

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Setting $f(x) = x^2 - 4 = 0$ gives zeroes $\alpha = 2$ and $\beta = -2$. We know:

Formula Used: $\alpha^4 + \beta^4 = (2)^4 + (-2)^4$

Evaluating directly: $2^4 + (-2)^4 = 16 + 16 = \boxed{32}$. [1M]

Q5. Assertion-Reason: [1]

Assertion (A): The polynomial $f(x) = x^2 - 2x - 3$ has zeroes 3 and -1.

Reason (R): Zeroes of a polynomial $p(x)$ are the x-coordinates of the points where $y = p(x)$ crosses the x-axis.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) Both A and R are true, but R is not the correct explanation of A.

We know both Factorisation and the Geometric Meaning of zeroes:

Formula Used: $f(x) = (x-3)(x+1) = 0 \implies x = 3, \ -1$

Checking Assertion **(A)**: $f(3) = 9 - 6 - 3 = 0$ and $f(-1) = 1 + 2 - 3 = 0$, so **(A)** is **true**. Reason **(R)** correctly states the general graphical definition of real zeroes, so **(R)** is **true**; however, **(R)** does not explain the algebraic factorisation yielding 3 and -1. Thus option $\boxed{\text{(B)}}$ is correct. [1M]

SECTION B: Very Short Answer Questions

MATHS MANTHAN | $3 \times 2 = 6$ Marks

Q6. If the sum of the zeroes of the quadratic polynomial $f(x) = kx^2 - 3x + 5$ is 1, find the value of k. [2]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Comparing $f(x) = kx^2 - 3x + 5$ $(k \neq 0)$ with $ax^2 + bx + c$, we have $a = k$, $b = -3$. We know:

Formula Used: Sum of zeroes $= -\dfrac{b}{a}$

Setting sum equal to 1:

$$-\frac{(-3)}{k} = 1 \implies \frac{3}{k} = 1 \implies \boxed{k = 3}. \qquad [2M]$$

Q7. If one zero of the polynomial $f(x) = 3x^2 + 8x + k$ is reciprocal of the other, find the value of k. [2]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let the zeroes be α and $\frac{1}{\alpha}$. We know Vieta's product formula:

Formula Used: $\alpha\beta = \dfrac{c}{a}$

Comparing $3x^2 + 8x + k$ $(a = 3, c = k)$:

$$\alpha \cdot \frac{1}{\alpha} = \frac{k}{3} \implies 1 = \frac{k}{3} \implies \boxed{k = 3}. \qquad [2M]$$

Q8. Find the zeroes of $p(t) = 5t^2 - 4 - 8t$ and verify the relationship between zeroes and coefficients. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 2 Marks]

Rewriting in standard form: $p(t) = 5t^2 - 8t - 4$ $(a = 5, b = -8, c = -4)$. We know:

Formula Used: $\alpha + \beta = -\dfrac{b}{a}, \qquad \alpha\beta = \dfrac{c}{a}$

Step 1 – Factorisation: Splitting $-8t = -10t + 2t$:

$$p(t) = 5t^2 - 10t + 2t - 4 = 5t(t-2) + 2(t-2) = (5t+2)(t-2) \implies \alpha = \boxed{-\frac{2}{5}},\ \beta = \boxed{2}. \qquad [1M]$$

Step 2 – Verification:

$$\alpha + \beta = -\frac{2}{5} + 2 = \frac{8}{5} = -\frac{(-8)}{5} = -\frac{b}{a}, \qquad \alpha\beta = \left(-\frac{2}{5}\right)(2) = -\frac{4}{5} = \frac{c}{a}. \qquad [1M]$$

SECTION C: Short Answer Questions

MATHS MANTHAN | $3 \times 3 = 9$ Marks

Q9. If α and β are zeroes of the quadratic polynomial $f(x) = x^2 - x - 4$, find the value of $\dfrac{1}{\alpha} + \dfrac{1}{\beta} - \alpha\beta$. [3]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

From $f(x) = x^2 - x - 4$, $a = 1$, $b = -1$, $c = -4$. We know Vieta's relations:

Formula Used: $\alpha + \beta = -\dfrac{b}{a} = 1, \qquad \alpha\beta = \dfrac{c}{a} = -4$

Simplifying $\frac{1}{\alpha} + \frac{1}{\beta} - \alpha\beta$:

$$\frac{1}{\alpha} + \frac{1}{\beta} - \alpha\beta = \frac{\alpha+\beta}{\alpha\beta} - \alpha\beta \qquad [1.5M]$$

$$= \frac{1}{-4} - (-4) = -\frac{1}{4} + 4 = \frac{-1+16}{4} = \boxed{\frac{15}{4}}. \qquad [1.5M]$$

Q10. If α and β are the zeroes of the quadratic polynomial $f(x) = 3x^2 - 4x + 1$, find a quadratic polynomial whose zeroes are $\dfrac{\alpha^2}{\beta}$ and $\dfrac{\beta^2}{\alpha}$. [3]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

From $f(x) = 3x^2 - 4x + 1$, $\alpha + \beta = \frac{4}{3}$ and $\alpha\beta = \frac{1}{3}$. We know:

Formula Used: $g(x) = k\left(x^2 - S'x + P'\right), \qquad \alpha^3 + \beta^3 = (\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)$

First compute $\alpha^3 + \beta^3$:

$$\alpha^3 + \beta^3 = \left(\frac{4}{3}\right)^3 - 3\left(\frac{1}{3}\right)\left(\frac{4}{3}\right) = \frac{64}{27} - \frac{4}{3} = \frac{64-36}{27} = \frac{28}{27}. \qquad [1M]$$

Now compute the sum S' and product P' of $\frac{\alpha^2}{\beta}$ and $\frac{\beta^2}{\alpha}$:

$$S' = \frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} = \frac{\alpha^3+\beta^3}{\alpha\beta} = \frac{28/27}{1/3} = \frac{28}{9}, \qquad P' = \left(\frac{\alpha^2}{\beta}\right)\left(\frac{\beta^2}{\alpha}\right) = \alpha\beta = \frac{1}{3}. \qquad [1M]$$

Taking $k = 9$, $g(x) = 9\left(x^2 - \frac{28}{9}x + \frac{1}{3}\right) = \boxed{9x^2 - 28x + 3}$. [1M]

Q11. If α, β are zeroes of the polynomial $x^2 - 6x + a$ and $3\alpha + 2\beta = 16$, find the value of a. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

From $x^2 - 6x + a$, we know Vieta's formulas:

Formula Used: $\alpha + \beta = 6, \qquad \alpha\beta = a$

Solving $\alpha + \beta = 6$ (1) together with $3\alpha + 2\beta = 16$ (2): multiplying (1) by 2 gives $2\alpha + 2\beta = 12$. Subtracting from (2):

$$(3\alpha + 2\beta) - (2\alpha + 2\beta) = 16 - 12 \implies \alpha = 4 \implies \beta = 6 - 4 = 2. \qquad \textbf{[2M]}$$

Hence $a = \alpha\beta = (4)(2) = \boxed{8}$. **[1M]**

SECTION D: Long Answer / HOTS Questions — MATHS MANTHAN | 2 × 5 = 10 Marks

Q12. If α and β are the zeroes of $p(x) = 3x^2 - 4x + 1$, form a quadratic polynomial whose zeroes are $\dfrac{\alpha^2}{\beta}$ and $\dfrac{\beta^2}{\alpha}$. Also, evaluate the expression $\alpha^4\beta^2 + \alpha^2\beta^4$. **[5]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

From $p(x) = 3x^2 - 4x + 1$, $\alpha + \beta = \frac{4}{3}$ and $\alpha\beta = \frac{1}{3}$. **[0.5M]**

We know the construction and symmetric power identities:

Formula Used: $g(x) = k\left(x^2 - S'x + P'\right), \quad \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta, \quad \alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$

Part 1 – Polynomial construction:

$$\alpha^3 + \beta^3 = \left(\frac{4}{3}\right)^3 - 3\left(\frac{1}{3}\right)\left(\frac{4}{3}\right) = \frac{64}{27} - \frac{4}{3} = \frac{28}{27}, \qquad \textbf{[1M]}$$

$$S' = \frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} = \frac{\alpha^3 + \beta^3}{\alpha\beta} = \frac{28/27}{1/3} = \frac{28}{9}, \qquad P' = \left(\frac{\alpha^2}{\beta}\right)\left(\frac{\beta^2}{\alpha}\right) = \alpha\beta = \frac{1}{3}. \qquad \textbf{[1M]}$$

Choosing $k = 9$ yields $\boxed{g(x) = 9x^2 - 28x + 3}$. **[0.5M]**

Part 2 – Evaluating $\alpha^4\beta^2 + \alpha^2\beta^4$: Factoring out $\alpha^2\beta^2 = (\alpha\beta)^2$:

$$\alpha^4\beta^2 + \alpha^2\beta^4 = (\alpha\beta)^2(\alpha^2 + \beta^2) = (\alpha\beta)^2[(\alpha + \beta)^2 - 2\alpha\beta] \qquad \textbf{[1M]}$$

$$= \left(\frac{1}{3}\right)^2\left[\left(\frac{4}{3}\right)^2 - 2\left(\frac{1}{3}\right)\right] = \frac{1}{9}\left(\frac{16}{9} - \frac{6}{9}\right) = \frac{1}{9} \times \frac{10}{9} = \boxed{\frac{10}{81}}. \qquad \textbf{[1M]}$$

Q13. A quadratic polynomial $f(x) = ax^2 + bx + c$ has zeroes α and β. **(i)** Prove that $\frac{1}{\alpha} + \frac{1}{\beta} = -\frac{b}{c}$. **(ii)** If $c = a$, prove that the zeroes are reciprocals of each other. **(iii)** If $b = 0$, prove that the zeroes are equal in magnitude and opposite in sign. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

For $f(x) = ax^2 + bx + c$ $(a \neq 0)$, we know Vieta's relations between zeroes and coefficients:

Formula Used: $\alpha + \beta = -\frac{b}{a}, \qquad \alpha\beta = \frac{c}{a}$

(i) Proof of $\frac{1}{\alpha} + \frac{1}{\beta} = -\frac{b}{c}$ $(c \neq 0)$: Taking common denominator $\alpha\beta$:

$$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{-b/a}{c/a} = -\frac{b}{a} \times \frac{a}{c} = \boxed{-\frac{b}{c}}. \qquad \textbf{[2M]}$$

(ii) Proof when $c = a$: Substituting $c = a$ into the product relation $\alpha\beta = \frac{c}{a}$:

$$\alpha\beta = \frac{a}{a} = 1 \implies \boxed{\beta = \frac{1}{\alpha}}$$ (i.e., the zeroes are reciprocals of each other). **[1.5M]**

(iii) Proof when $b = 0$: Substituting $b = 0$ into the sum relation $\alpha + \beta = -\frac{b}{a}$:

$$\alpha + \beta = -\frac{0}{a} = 0 \implies \boxed{\beta = -\alpha}$$ (equal in magnitude and opposite in sign). **[1.5M]**

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